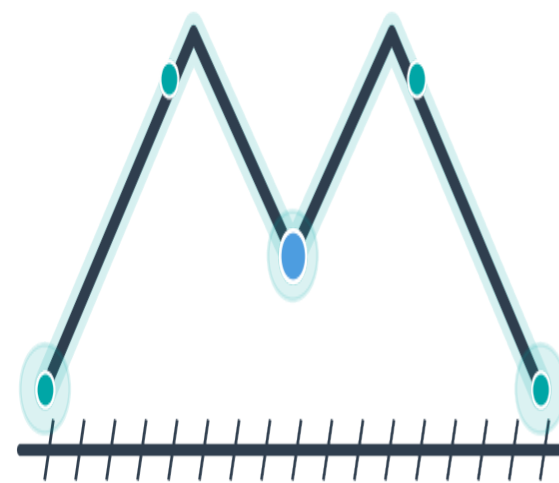




智轨百年

A CENTURY ON THE SMART RAIL

JING-ZHANG SMART RAIL · HAIDIAN

A Century on the Smart Rail

Urban Design for the Centennial Jing-Zhang AI Innovation Belt

JING-ZHANG SMART RAIL · HAIDIAN

v0.3 · 2026-08-08 · 百年京张AI创新带城市设计开源征集 / Open City · Haidian

Status: AI-generated conceptual design. All spatial ideas are conceptual suggestions / reference schemes for professional teams — not statutory planning, approval, engineering conclusions or investment commitments. Official precise redlines are missing; provisional repo boundary used; full recalculation upon official data release.

01 Overview and three-level scope

Research 43.6 km² / design 11.4 km² / key areas 368.4 ha; submitted boundary recomputed 11,412,842 m², key areas 3 (3,692,893 m²).

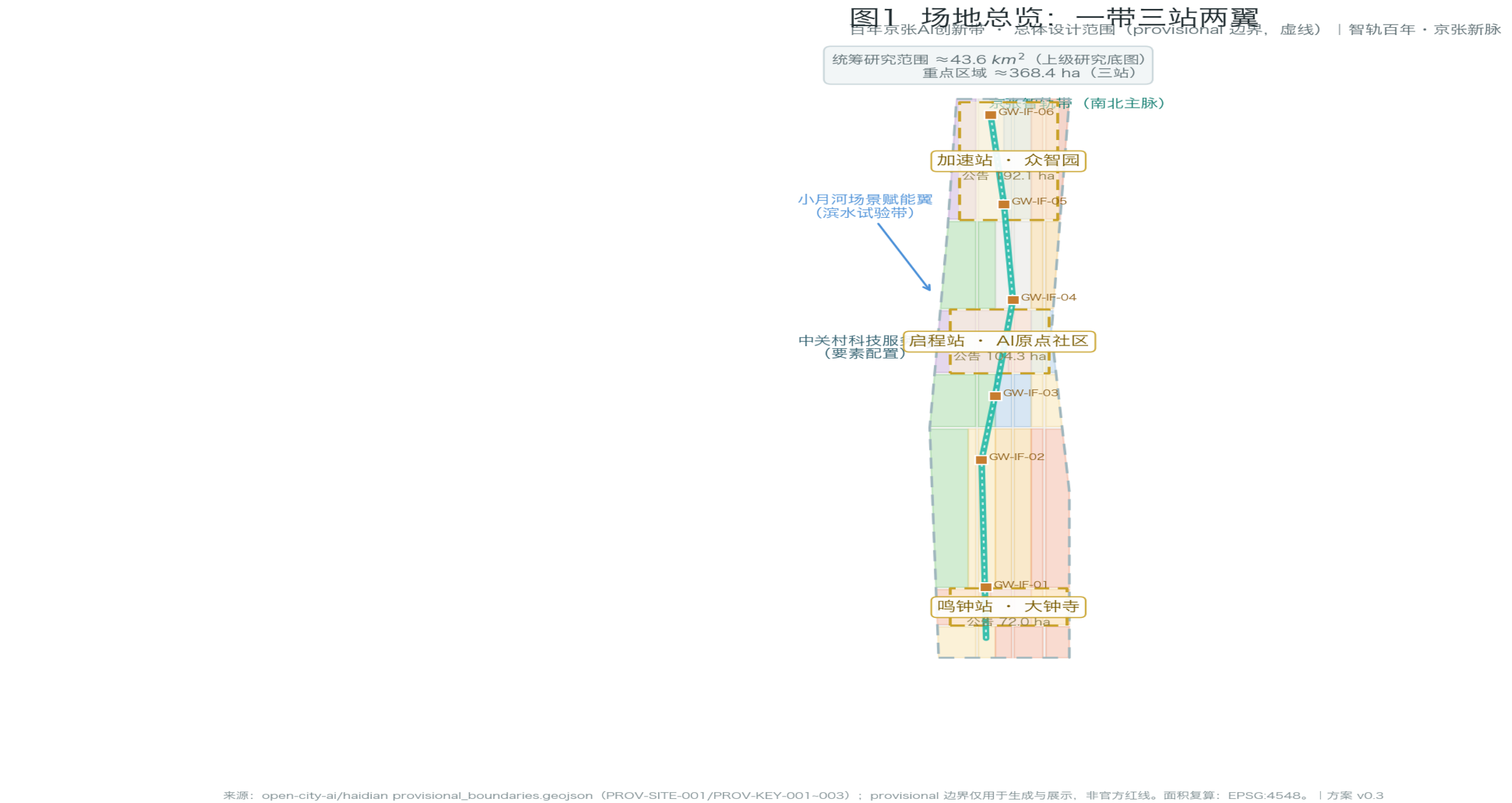


Fig: proposal derived figures (from submitted GeoJSON/metrics)

Site / m ²	Road km	Green	Public	Road
11,412,842	None	18.7%	1.9%	8.9%

02 Core concept: one track, a century relay

1909 Zhan Tianyou built the Jing-Zhang Railway; 2026 the same corridor hosts an AI innovation belt. Railway semantics (station/track/corridor/crossing/timetable) are translated into spatial grammar: master name Jing-Zhang Smart Rail, 4-level naming (belt—station—corridor—node), logo as zigzag 'zhi' motif (ren-shaped track + letter Z + data pulse), JZ blue / smart-rail light palette.

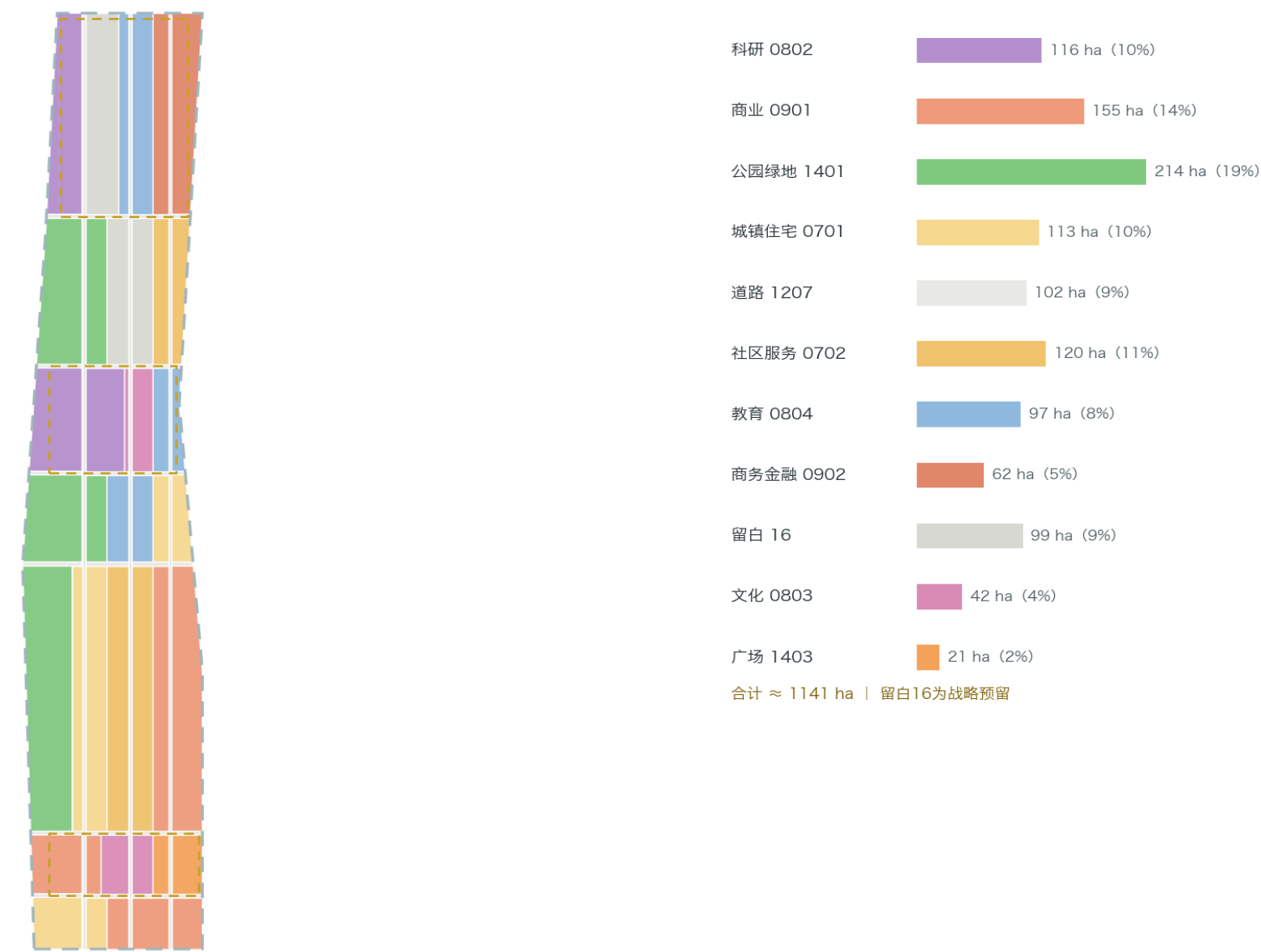
京张铁路（1909）	AI创新带（2026）
钢轨与道床	算力网络+数据通道（智轨光廊）
车站	三大重点片区（三站）
机务段/编组场	众智园：训练/评测/加速
始发站	AI原点社区：思想与人才出发
到站/货场	大钟寺：AI进入城市生活与消费
列车时刻表	年度活动日历与场景运营节奏
铁路工程师	AI开发者/研究者/创业者
詹天佑精神	全栈自主创新与开源开放精神

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03 Structure: one belt, three stations, two wings

Smart Rail belt north-south; Origin Station (ideas), Acceleration Station (Zhongzhiyuan), Bell Station (Dazhongsi); Zhongguancun service wing (factors), Xiaoyuehe scenario wing (validation). Five functions explicitly mapped to spatial carriers.

图2 用地结构：概念分区与账本



来源：open-city-ai/haidian provisional_boundaries.geojson (PROV-SITE-001/PROV-KEY-001~003) ； provisional 边界仅用于生成与展示，非官方红线。面积复算：EPSG:4548。 | 方案 v0.3

Fig: proposal derived figures (from submitted GeoJSON/metrics)

Site / m ² 11,412,842	Road km None	Green 18.7%	Public 1.9%	Road 8.9%
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04 Key areas: three stations

Origin Community 104.3 ha (Origin Stele, scholar-developer mixed blocks); Zhongzhiyuan 192.1 ha (compute/eval clusters, marshalling yard, governance plaza, green ring); Dazhongsi 72.0 ha (Bell of Intelligence, smart-native commerce, gateway plaza). Conceptual; intensity pending official controls.

五大功能 / Five functions	空间载体 / Spatial carrier
	Zhongzhiyuan: compute/eval clusters, marshalling yard, green compute
World-class AI ecosystem	Origin: mixed blocks, open-source market, collaboration stations
	12 passports + Departure Protocol (3 industry-test cards)
Intelligent vibrant city	Smart Rail corridor, smart-native commerce, waterfront strip
	Governance plaza, Bell Witness, model-card registry, forum

图3 重点片区详设结构（三站，provisional 范围）

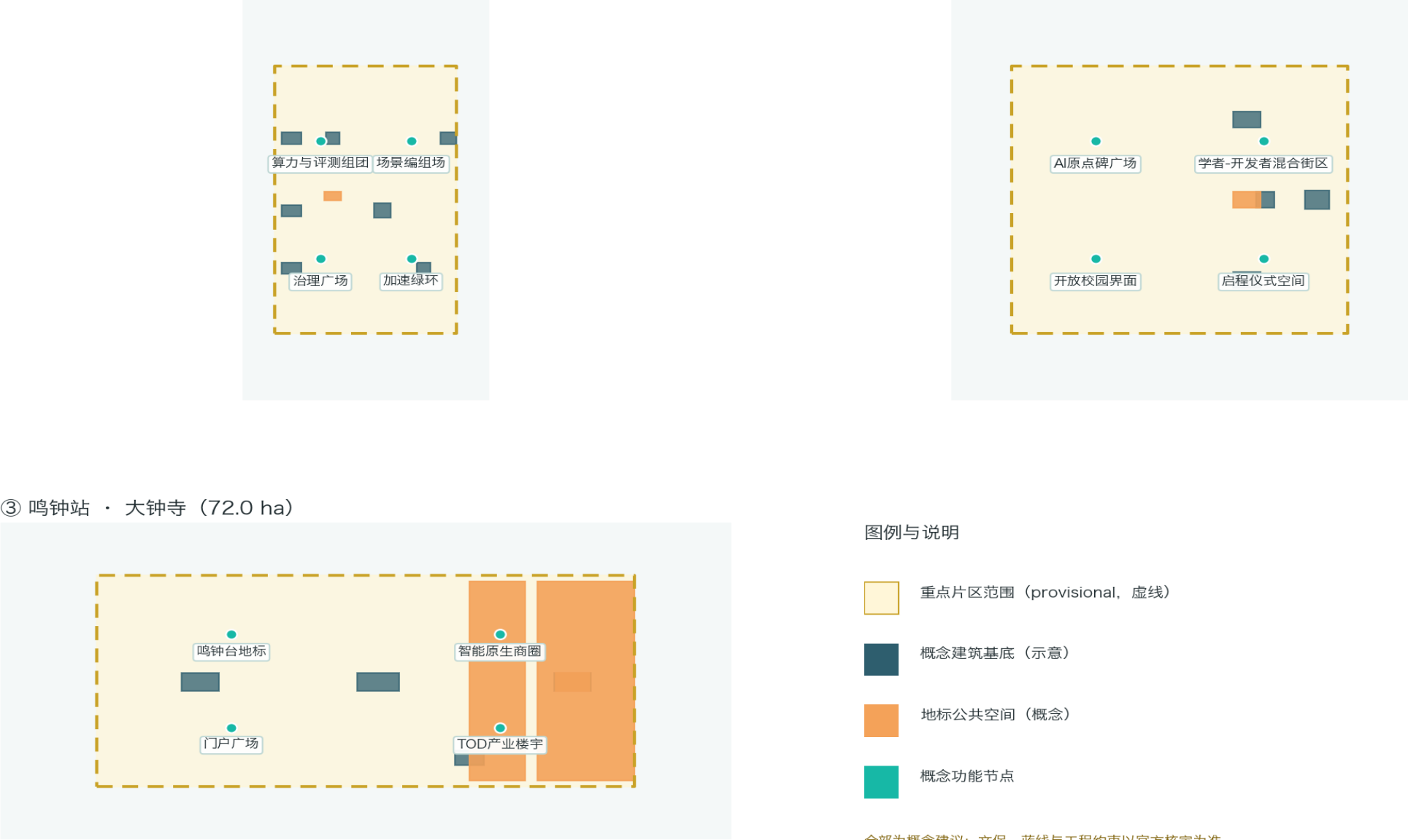


Fig: proposal derived figures (from submitted GeoJSON/metrics)

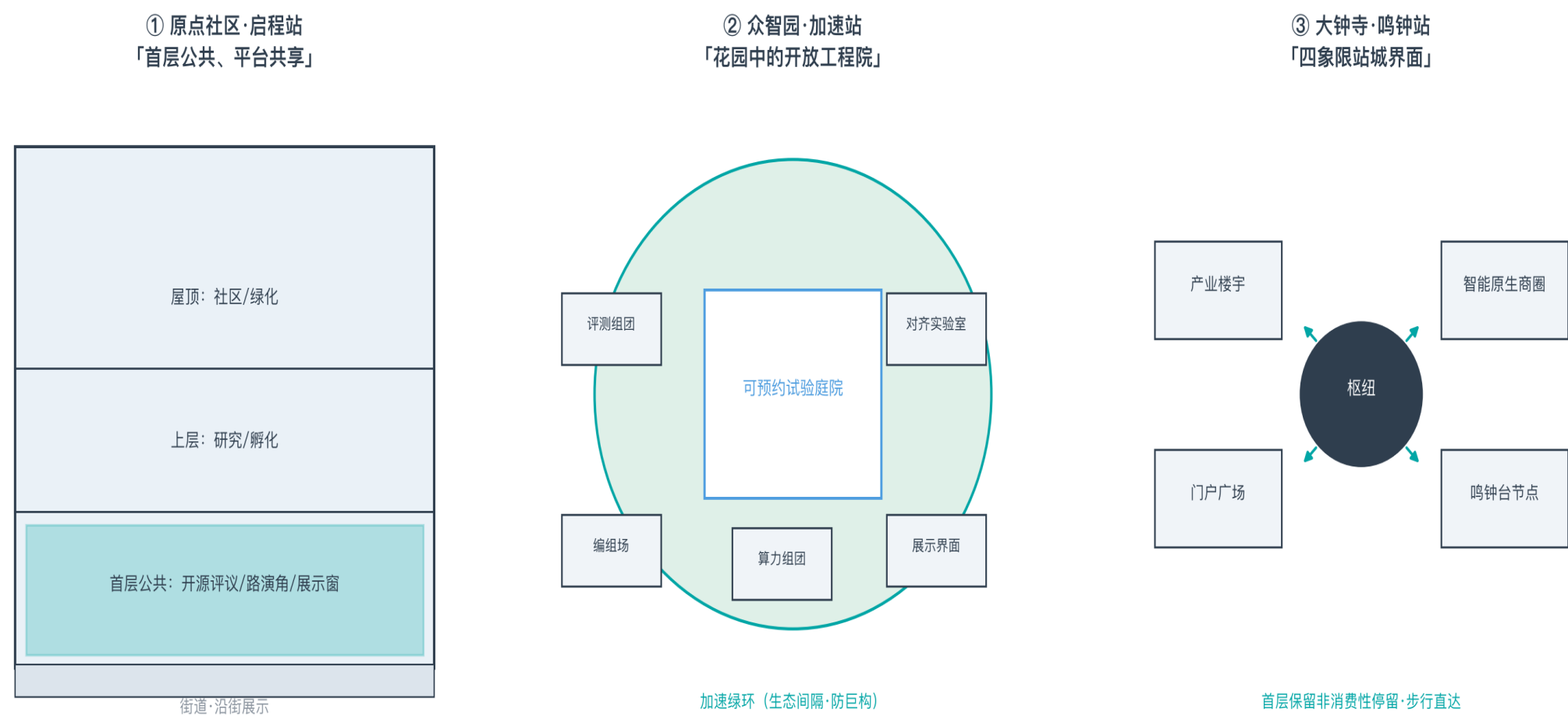


Fig: three spatial prototypes (conceptual) — public ground floor / open engineering yard / four-quadrant station-city interface

Site / m ²	Road km	Green	Public	Road
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05 AI scenarios: 12 scenario passports

12 scenario cards (incl. 3 industry-test ★) upgraded to Scenario Passports: spatial mapping/users/data/privacy/human review/operator/risks-exit + admission/RACI/KPI/stop threshold/appeal/data retention.

卡/准入	RACI	KPI	停止阈值	申诉与撤回	数据期限	
S01 AI tutoring block	Institution-led + parental consent	School/district/teachers	Term satisfaction	Privacy incident → stop	Parent & school	Deleted end of term
S02 Open-source market	Open registration + OSS license	Committee + operator	Projects/conversions	Order risk → pause	Organizer	No personal data
S03 Researcher station	Inter-institutional agreement	University alliance + operator	Utilization/outputs	Data violation → stop	Institutional appeal	Destroyed per agreement
S04 LLM eval sandbox ★	3rd-party + red-team vetting	Evaluator/governance/safety officer	Reports/fix rates	Major safety event → stop	Public appeal + review	Isolated retention
S05 Robotics proving ground ★	Enclosed + safety assessment + insurance	Operator/regulator/emergency	Success/incident rates	Incident → stop + postmortem	On-site + written	Anonymized sensors
S06 Autonomous shuttle ★	Limits + sandbox permit + attendant	Operator/traffic police/attendant	Ridership/takeover rate	Takeover over limit → stop	Passenger + public	Trip data deleted
S07 Dispatch center	Open application + review trail	Operator + 3rd-party review	Scenarios/graduation	Review violation → pause	Appeal + review	Limited-term retention
S08 Smart-native district	Merchant compliance + consent	District operator + regulator	Experience/complaints	Data violation → stop	Consumer channel	Minimal consumption data
S09 Urban vital signs	Statistical caliber + anonymization	Data office + 3rd-party audit	Coverage/accuracy	Anonymization failure → stop	Data objection	Statistics only
S10 Cultural AI guide	Content review + expert check	Culture dept + operator	Usage/corrections	Factual error → freeze	Visitor feedback	No location stored
S11 Waterfront test strip	Ecological review + device disclosure	Water/subdistrict/operator	Pilots/eco impact	Ecological impact → stop	Resident feedback	Anonymized sensing
S12 Community health station	Licensed + voluntary	Medical/subdistrict/practitioners	Service count/satisfaction	Medical incident → stop	Resident + patient	Minimal retention

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06 Regional synergy matrix

Beiwai community, Future Science City, Huairou Science City, E-Town, Jing-Jin-Ji: capability—interface—data/IP boundary interface list.

协同对象	能力互补	接口与要素流	数据/IP边界
Beiwai community	Youth living & community trials	Xiaoyuehe wing north extension	No personal data sharing
Future Science City	Basic research & frontier facilities	Zhongzhiyuan eval/compute interface	Data stays in-domain
Huairou Science City	Big-science facilities & original innovation	Origin research corridor	Outputs by agreement
E-Town	Intelligent manufacturing & landing	Dazhongsi southbound uptake	Authorized use
Jing-Jin-Ji	Regional talent/scenarios/markets	Belt annual-event radiation	Public outputs reusable

Site / m ²	Road km	Green	Public	Road
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07 Operation protocol: Departure Protocol

Issue → Review → Sandbox → Bell Witness → Depart → Postmortem, plus Rollback; Human Override at every step.

- 1. Issue: public submission, full trail, appealable
- 2. Review: Origin Community — provenance/idea/public-interest gate
- 3. Sandbox: Zhongzhiyuan — controlled validation, admission & stop thresholds
- 4. Bell Witness: Dazhongsi — public witnessing & honor records
- 5. Depart: into real city scenarios, time/area limited
- 6. Postmortem: Xiaoyuehe back-testing, feedback returns
- 7. Rollback: pause, withdrawal & site restoration

Site / m ²	Road km	Green	Public	Road
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08 Implementation governance and risk response

P1-P7 entry/exit gates, named human-review roles, P0/P1/P2 response targets, conditional cost envelope, model cards/algorithm registry; three perceptible AI city-form changes (buildings/streets/governance).

P0/P1/P2 响应目标：P0 人身/重大安全事故——立即暂停+15分钟响应+24小时复盘；P1 数据/隐私事件——4小时处置并通知；P2 一般投诉——48小时受理留痕。条件成本包络：低成本可撤回试点→中量级示范段→全域网络，不编造投资数字。三类可感知变化：建筑（可预约共享实验/开源评议/产业交换）、街道（智轨慢行主干+东西缝合道口+无障碍人工等价通道）、治理（场景护照+停止阈值+拒绝清单+失败档案）。

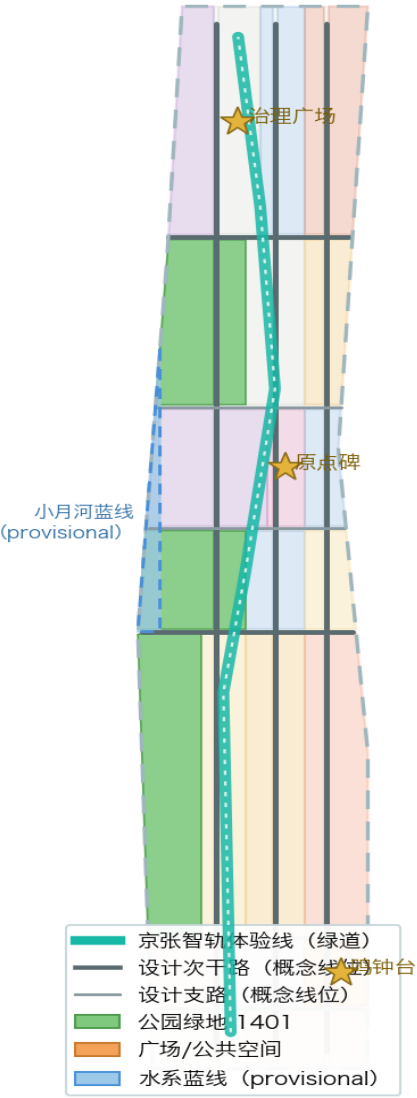
优先级	事件类型	响应目标（概念级）
P0	人身/重大安全事故	立即暂停 + 15分钟响应 + 24小时复盘
P1	数据/隐私事件	4小时处置并通知受影响者
P2	一般投诉与申诉	48小时受理并留痕
——	具名人工复核角色	规划专业/交通专项/文物部门/数据安全合规/社区代表/第三方评审

Site / m ² 11,412,842	Road km None	Green 18.7%	Public 1.9%	Road 8.9%
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09 Land use, mobility, blue-green

Research 1,163,124 m ² , commercial unknown m ² , residential 1,133,054 m ² , roads 1,016,948 m ² (8.9%), green 2,139,223 m ² (18.7%), public space 219,202 m ² (1.9%).

图4 交通与蓝绿：路网 · 智轨体验线 · 绿网 · 公共空间



来源：open-city-ai/haidian provisional_boundaries.geojson（PROV-SITE-001/PROV-KEY-001~003）；provisional 边界仅用于生成与展示，非官方红线。面积复算：EPSG:4548。| 方案 v0.3

Fig: proposal derived figures (from submitted GeoJSON/metrics)

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10 Phasing and renewal projects

Phase 1 ignite (2026-27, 2,370,155 m ²) Origin demo; Phase 2 form (2028-29, 5,861,693 m ²) Zhongzhiyuan + full line; Phase 3 mature (2030-33, 3,180,994 m ²) Dazhongsi and belt-wide network. Projects P1-P7 all with entry/exit gates.

- P1 Origin Stele Plaza & departure ritual space — — Entry: heritage & site permits; exit: veto/structural risk
- P2 Smart Rail experience-line demo segment — — Entry: slow-mobility & safety assessment; exit: incident → stop
- P3 Open-source market permanent carrier — — Entry: operator + site agreement; exit: order risk → pause
- P4 Eval sandbox & dispatch center — — Entry: safety assessment + 3rd-party review; exit: major event → stop
- P5 Bell of Intelligence & gateway plaza — — Entry: heritage authority review; exit: heritage objection
- P6 Smart-native commerce interface retrofit — — Entry: merchant compliance + consent; exit: data violation → stop
- P7 Waterfront scenario test strip — — Entry: ecological avoidance review; exit: eco impact → stop

阶段	面积	重点动作
Phase 1 Ignite 2026-27	2,370,155 m ²	Origin demo: stele, market, first scenario pilot
Phase 2 Form 2028-29	5,861,693 m ²	Full line + Zhongzhiyuan: sandbox/proving-ground operations
Phase 3 Mature 2030-33	3,180,994 m ²	Dazhongsi bell segment and belt-wide network

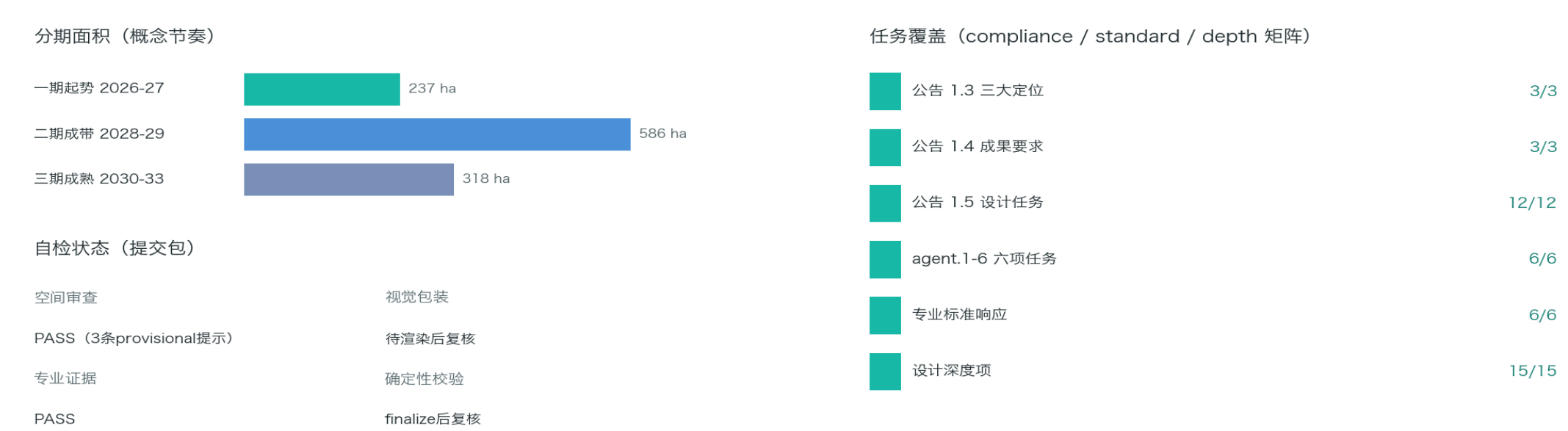
Site / m ² 11,412,842	Road km None	Green 18.7%	Public 1.9%	Road 8.9%
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11 Metrics evidence and compliance

All metrics recomputed EPSG:4548; compliance_matrix 23 entries (1.3.1-1.5.3.3) + agent.1-6; standard_matrix 6 standards; design_depth_matrix 15 items complete; four self-check gates PASS.

图5 指标证据：核心指标复算 · 分期 · 任务覆盖 · 自检状态

场地面积 11.41 <i>km</i> ² 总体设计范围 (provisional复算)	商业用地0901 155 ha 沿枢纽与门户	重点片区 369.3 ha 三站合计	公园绿地 18.7% 214 ha
公共空间 1.9% 22 ha	道路用地 8.9% 102 ha	科研用地 116 ha 0802	容积率/高度 待官方确认 planning_limits 缺失项



来源：open-city-ai/haidian provisional_boundaries.geojson（PROV-SITE-001/PROV-KEY-001-003）；provisional 边界仅用于生成与展示，非官方红线。面积复算：EPSG:4548。| 方案 v0.3 指标与 metrics.json 一致，可被空间审查复算。

Fig: proposal derived figures (from submitted GeoJSON/metrics)

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12 Risk, copyright and self-check

8-dimension risk matrix (privacy/complexity/acceptance/cost/policy/spatial/technology/equity) with mitigation and human review; per-asset rights ledger; four local gates PASS (provisional notes minor).

风险维度	缓解与人工复核
Data privacy	Minimization, anonymized aggregation, licensed health review
Implementation complexity	Lightweight reversible pilots first, phased
Public acceptance	Disclosure, appeal channels, opt-out/detour
Operations cost	Reuse existing operators, phased evaluation
Policy uncertainty	All marked pending confirmation, no numeric promises
Spatial dispute	Time/zone separation, reversible kit, heritage/blue-line buffers
Technology maturity	Controlled test + regulatory sandbox, human takeover
Equity & inclusion	Accessibility, equivalent human channels, non-digital alternatives

Site / m ²	Road km	Green	Public	Road
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